

# Mu-Ko (Claire) Chou

✉ nm6144012@gs.ncku.edu.tw    🌐 claire891018.github.io

## RESEARCH INTERESTS

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Mechanistic Interpretability; Multilingual Legal NLP

## EDUCATION

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**National Cheng Kung University**, Tainan, Taiwan Feb. 2025 – Expected Feb. 2027  
M.S. in Intelligent Technology Systems, Miin Wu School of Computing GPA: 4.19/4.30  
Advisor: Prof. Shao-Man Lee

**National Cheng Kung University**, Tainan, Taiwan Sep. 2019 – Jun. 2024  
B.A. in Political Science, College of Social Sciences

**Technical University of Darmstadt**, Darmstadt, Germany Spring 2023  
Exchange Student

## RESEARCH EXPERIENCE

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**National Cheng Kung University**, Tainan, Taiwan Advisor: Prof. Shao-Man Lee  
*SocioLegal Lab, Miin Wu School of Computing*

**M.S. Thesis Research** Feb. 2026 – Present

**Research:** *Mechanistic Interpretability of Multilingual Legal Concepts*

- Investigate whether multilingual large language models encode shared representations of constitutional concepts using sparse autoencoders (SAEs).
- Developed a cross-lingual analysis pipeline over 8,654 constitutional judgments in 25 languages, identifying concept-associated SAE features and tracing their activations back to source passages.
- Resulted in *Tracing Constitutional Concepts Across Languages with Sparse Autoencoder Features*, currently under review at NLLP 2026.

### Part-Time Research Assistant

**Research:** *Retrieval-Augmented Generation and Evaluation for Taiwanese Legal Documents* Aug. 2025 – Present

- Develop TaDELS-RAG, a retrieval-augmented generation system and evaluation benchmark for source-grounded question answering over Taiwanese legal documents.
- Co-developed a benchmark of 211 legal documents and 648 human-reviewed question-answer pairs, together with an entity-oriented evaluation framework for answer faithfulness and hallucination.
- Resulted in the co-first-authored *TaDELS-RAG: Entity-Oriented Evaluation Dataset for Retrieval-Augmented Generation in Taiwanese Law*, presented at the 2026 Empirical Legal Studies Workshop.

**Research:** *Intersectional and Cross-Cultural Biases in Large Language Models* 2024 – 2025

- Investigated intersectional and cross-cultural biases in LLM moral decision-making across demographic and cultural profiles.
- Evaluated 5 LLMs across 1,769 demographic profiles, 9 cultural clusters, and 38,350 Moral Machine decisions using Average Marginal Component Effects (AMCE) and human-model distance measures.
- Analyzed interaction effects across demographic attributes; findings were presented by Prof. Shao-Man Lee at the Young Scholar Workshop on Empirical Legal Studies in the Sinophone Region (2025).

**Full-Time Research Assistant** Jul. 2024 – Jan. 2025

**Government Project:** *AI-Assisted Prosecution System, Ministry of Justice (Taiwan)*

- Served as Technical Project Manager for a government legal-AI project, coordinating requirements, implementation, testing, and communication between prosecutors, government stakeholders, and the technical team.
- Contributed hands-on to NLP-based processing of investigative records and AI-assisted document workflows, including extraction and validation of key case information from police and interrogation records.

## PUBLICATIONS & PRESENTATIONS

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### Manuscripts Under Review

*Tracing Constitutional Concepts Across Languages with Sparse Autoencoder Features.*

Shao-Man Lee, **Mu-Ko Chou**

Under review at the Natural Legal Language Processing (NLLP) Workshop 2026.

### Workshop Papers & Presentations

*TaDELS-RAG: An Entity-Oriented Evaluation Dataset for Retrieval-Augmented Generation in Taiwanese Law.*

Shao-Man Lee\*, **Mu-Ko Chou\***, Chun-Che Huang, Chin-You Hou

Empirical Legal Studies Workshop 2026, Aug. 5–6, 2026.

\*Equal contribution.

*Navigating Judicial Overload: A Multi-Agent System for Argument Tracking in Constitutional Courts.*

Shao-Man Lee, **Mu-Ko Chou**, Yu-Chiao Chen, Chun-Che Huang, Chun-Chieh Niu, Chin-You Hou

Computational Legal Studies Workshop 2025, Singapore Management University, Sep. 8–9, 2025.

INDUSTRY EXPERIENCE

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Cathay Financial Holdings, Kaohsiung, Taiwan  
Cathay Development Center (CDC)

Feb. 2024 – Jun. 2024

Local Community Engagement Intern

- Developed and evaluated a GCP-based RAG system for financial regulatory knowledge retrieval, comparing multiple retrieval and chaining strategies with context precision, context recall, and answer relevancy.
- Implemented PostgreSQL data structures and source-attribution mechanisms to improve response traceability and reduce hallucinations.

HONORS AND AWARDS

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The Phi Tau Phi Scholastic Honor Society, Honorary Member

2026

Outstanding Student Scholarship, Miin Wu School of Computing, NCKU

2025 – 2026

Competitive scholarship for top-ranked master’s admits; renewed for a second year based on academic and research performance.

SELECTED COURSEWORK

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Natural Language Processing; Advanced Legal Large Language Model Implementation; Fundamentals of Deep Learning Networks: Theory and Industrial Applications; Singular Value Decompositions and Data Analysis; Artificial Intelligence and Democratic Governance; Computer Vision and Deep Learning

SKILLS

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**Programming:** Python, SQL, LaTeX

**ML/NLP:** PyTorch, Hugging Face Transformers, Retrieval-Augmented Generation (RAG), Sparse Autoencoders (SAEs)

**Tools:** Git, PostgreSQL, Google Cloud Platform (GCP)